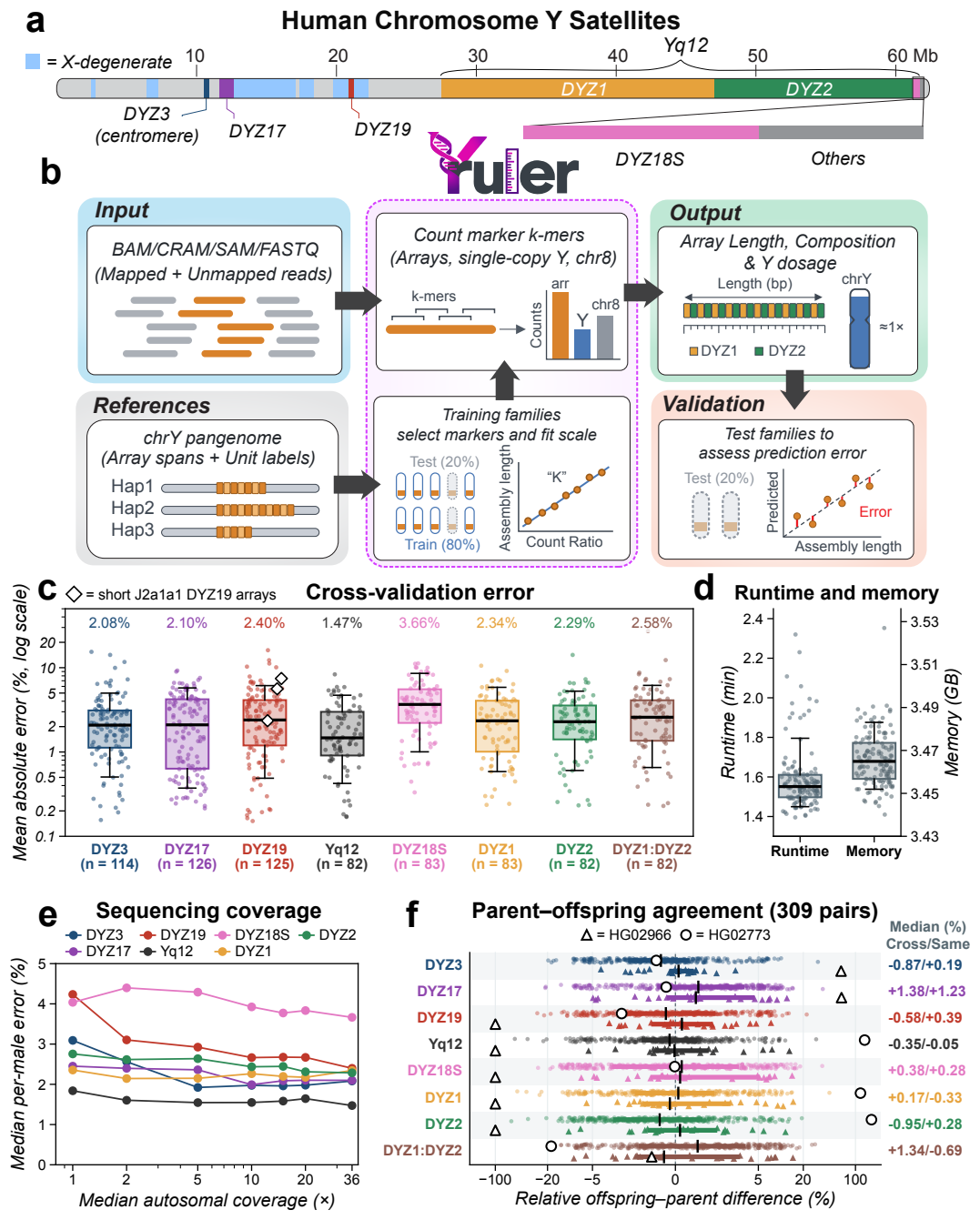

a
Human Chromosome Y Satellites
= X-degenerate
10
20
30
40
Yq12
50
60 Mb
DYZ1
DYZ2
DYZ3
(centromere)
DYZ17
DYZ19
DYZ18S
Others
b
Yuler
Input
BAM/CRAM/SAM/FASTQ
(Mapped + Unmapped reads)
References
chrY pangenome
(Array spans + Unit labels)
Hap1
Hap2
Hap3
Count marker k-mers
(Arrays, single-copy Y, chr8)
k-mers
Counts
arr
Y
chr8
Training families
select markers and fit scale
Test (20%)
Train (80%)
Assembly length
"K"
Count Ratio
Output
Array Length, Composition
& Y dosage
Length (bp)
chrY
≈1×
DYZ1
DYZ2
Validation
Test families to
assess prediction error
Test (20%)
Predicted
Error
Assembly length
c
◇ = short J2a1a1 DYZ19 arrays
Cross-validation error
Mean absolute error (%, log scale)
2.08%
2.10%
2.40%
1.47%
3.66%
2.34%
2.29%
2.58%
DYZ3 (n = 114)
DYZ17 (n = 126)
DYZ19 (n = 125)
Yq12 (n = 82)
DYZ18S (n = 83)
DYZ1 (n = 83)
DYZ2 (n = 82)
DYZ1:DYZ2 (n = 82)
d
Runtime and memory
Runtime (min)
Memory (GB)
Runtime
Memory
e
Sequencing coverage
DYZ3
DYZ19
DYZ18S
DYZ2
DYZ17
Yq12
DYZ1
Median per-male error (%)
Median autosomal coverage (×)
f
Parent–offspring agreement (309 pairs)
△ = HG02966 ○ = HG02773
Median (%)
Cross/Same
DYZ3 -0.87/+0.19
DYZ17 +1.38/+1.23
DYZ19 -0.58/+0.39
Yq12 -0.35/-0.05
DYZ18S +0.38/+0.28
DYZ1 +0.17/-0.33
DYZ2 -0.95/+0.28
DYZ1:DYZ2 +1.34/-0.69
Relative offspring–parent difference (%)